

State the domain and range of a function

Pre-Cal | CK-12: Domain and Range

Key Idea

Domain is every allowed input x ; **range** is every output y . Two things restrict a domain: you cannot divide by 0, and you cannot take the square root of a negative.

Key Terms

- **Domain** — allowed x -values
- **Range** — resulting y -values

Worked steps

Step 1: Set any denominator $\neq 0$; set any radicand ≥ 0 .

Step 2: Solve for x to get the domain.

Example: for $f(x) = \sqrt{x-2}$, need $x-2 \geq 0$, so $x \geq 2$.

You Try

Give the domain of $f(x) = \frac{5}{x+4}$.